Chhota Shigri Glacier Mapping with GlaViTU — Technical Report

Architecture · Error sources · Preprocessing · Analysis of all years (2017 baseline)

This report explains how the project maps Chhota Shigri Glacier (Himachal Pradesh, India) from
satellite data, and what the results show for every year from 2017 to 2025. It is written to be read
from start to end, but each section also stands on its own.

Every number in this report comes from the project's code, configuration, logs or output files. The source of each number is given in the text or in the file reference.

Item	Value
Glacier	Chhota Shigri, Chandra valley, Lahaul (`RGI60-14.15990` / `RGI2000-v7.0-G-14-30579`)
Satellite data	Sentinel-2 L1C (model input), Copernicus DEM GLO-30, Sentinel-1 RTC (M3 only)
Model	GlaViTU (Maslov et al., 2024), about 10 M parameters
Analysis grid	UTM 43N, 10 m, 1434 × 1254 px; analysis domain 47.9 km²
Years	2017–2025; odd years = analysis epochs, even years = M2 training years
Primary model	M2 (best held-out HMA IoU: 0.779)

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How to read this report

- **Sections 1–3** explain *how* the system works. **Section 4** shows *what* it found.
- Each section ends with a short summary table. If you are short of time, read these tables and 4.9 Key findings.
- Technical terms are explained in the glossary.
- Symbols used in tables:
    - **†** = a training year of model M2. These values are **not independent** (see 4.1).
    - **‡** = scenes that were **not co-registered**, so the uncertainty is underestimated.
    - ● addressed explicitly · ◐ partly or indirectly addressed · ○ not addressed.

Pipeline at a glance

The whole workflow runs from one script, `run_pipeline.sh`. No step uses a hand-drawn line.

```
 0. Get code and weights        third_party/glavitu (fixed commit),
pretrained weights + SHA-256 check
 1. Get study area and data     inventory → DEM → Sentinel-2 → co-
registration → Sentinel-1
 2. Get HMA benchmark tiles     expert-labelled training data, read by HTTP
byte ranges
 3. Train models                M1 (HMA) → calibration → pseudo-labels → M2
(site) ; M3 (SAR)
 4. Test models                 57 held-out HMA tiles, paired bootstrap
confidence intervals
```

```
 5. Make yearly outlines        multi-date ensemble per year → post-
processing → polygon
 6. Measure change              area, terminus, year-of-loss, spatial
patterns
 7. Validate site results       agreement with RGI7, published values,
temporal consistency
```

Two groups of years are kept apart on purpose:

Group	Years	Used for
Odd years	2017, 2019, 2021, 2023, 2025	Analysis only. No model is trained on them.
Even years	2018, 2020, 2022, 2024	Making pseudo-labels to train M2 only.

This split prevents M2 from learning the exact images in which change is measured.

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1. GlaViTU architecture

GlaViTU means **Gla**cier **Vi**sion **T**ransformer **U**-Net. It is a supervised deep-learning model for **semantic segmentation**: it gives every pixel a class (glacier or not glacier). All models in this project (M0, M1, M2, M3) use this same network. They differ only in their weights, training labels and input channels (see 1.10).

1.1 Source repository

Item	Value
Repository	https://github.com/konstantin-a-maslov/towards_global_glacier_mapping
Local folder	`third_party/glavitu/`
Fixed commit	`8c4eead670ef64764f8db2708bba3c8bbbfc67bb` (3 Jan 2025)
Paper	Maslov, Persello, Schellenberger & Stein (2024), *Globally scalable glacier mapping by deep learning matches expert delineation accuracy*, Nature Communications, doi:10.1038/s41467-024-54956-x
Licence	GNU GPL v2
Weights	Downloaded from the authors' Google Drive into `models/pretrained/`, checked with SHA-256

The commit is fixed so that later changes by the authors cannot change our results.

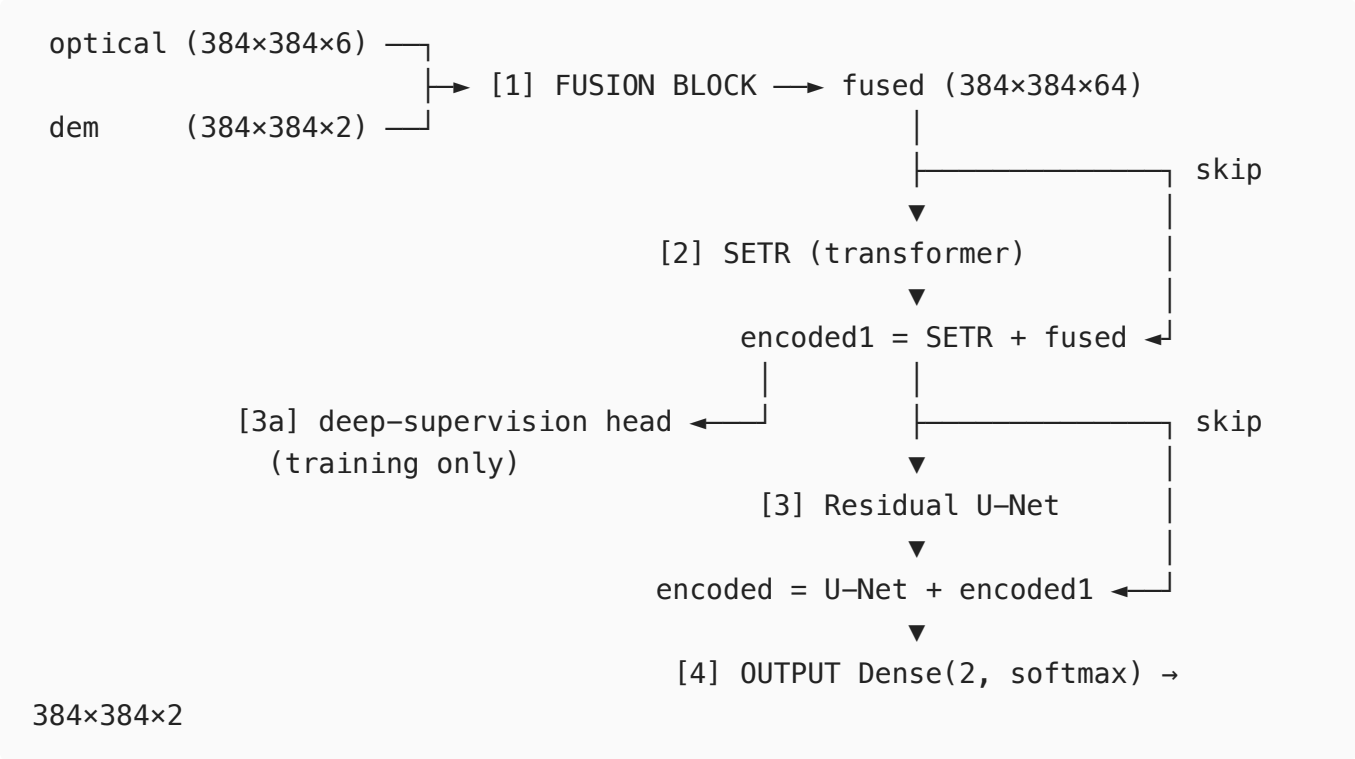
From this repository the project uses:

- the model code (`models/mapping/glavitu.py` , `setr.py` , `resunet.py` , and the `layers/` folder);
- the data loader, samplers, augmentations and focal loss (reused by `src/training/finetune.py`);
- the normalisation statistics `utils/dataset_stats.pickle` ;
- the sliding-window prediction logic of `deploy.py` , copied into `src/inference/glavitu.py` because `deploy.py` opens the training file when it is imported.

> **Licence note.** GPL v2 is a copyleft licence. Code that includes or modifies GlaViTU code and
> is distributed must also be under a GPL-compatible licence.

1.2 The network in one picture

Shapes are for one 384 × 384 px patch with the standard 8-channel input.



The idea in one sentence: the transformer looks at the whole patch to understand large shapes,
and the U-Net then makes the edges sharp at the full 10 m resolution.

1.3 Stage 1 — Fusion block

Code: `FusionBlock(64)` in `third_party/glavitu/layers/general/layers.py`.

Each input group (`optical` , `dem` , and for M3 `co_pol_sar`) has **its own branch**:

Step	Layer	Output (optical)	Purpose
1	Conv 3×3, 64 filters	384×384×64	Finds local patterns: edges, texture, colour combinations
2	BatchNorm + LeakyReLU	384×384×64	Stabilises values, adds non-linearity
3	Conv 3×3, 64 filters	384×384×64	Combines patterns into more complex ones
4	BatchNorm + LeakyReLU	384×384×64	As step 2
5	Global average pooling	1×1×64	One summary value per channel ("squeeze")
6	Dense 64 + LeakyReLU	1×1×64	Learns which channels matter for this patch
7	Dense 64 + sigmoid	1×1×64	A weight from 0 to 1 for each channel ("excitation")
8	Multiply step 4 × step 7	384×384×64	Strengthens useful channels, weakens others

Steps 5–8 form a **squeeze-and-excitation (SE)** block. In this implementation the SE hidden layer
is not reduced (64 → 64).

The branches are then **added**: `fused = optical_branch + dem_branch (+ sar_branch)` → 384×384×64.

Why separate branches? Reflectance, elevation and radar are different kinds of data. Each gets
its own feature extractor before they are combined. This also made M3 easy to build: a SAR branch
was added and all other layers were kept.

1.4 Stage 2 — SETR transformer

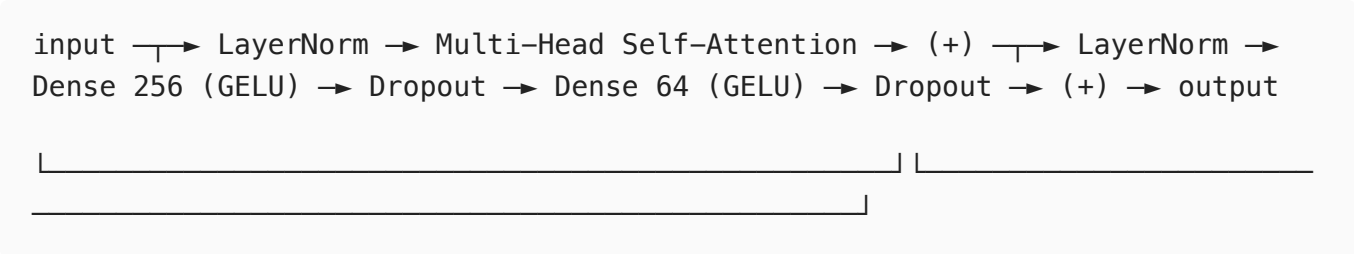
Code: `SETR(...)` in `models/mapping/setr.py` and `ViTEncoder` in `models/misc/vit.py`.
SETR means **SE**gmentation **TR**ansformer. GlaViTU uses a small version:

Setting	Value
Patch (token) size	16 × 16 px (160 m × 160 m)
Embedding size	64
MLP size	256
Transformer blocks	12
Attention heads	4 (16 dimensions each)

a) Turn the image into tokens

Step	Layer	Output	Purpose
1	Patch extraction 16×16	576 × 16,384	Cuts the patch into a 24 × 24 grid = 576 squares
2	Patch embedding (Dense 64)	576 × 64	Compresses each square into a 64-number token
3	Learned positional embedding	576 × 64	Tells the model where each square is

b) Transformer block, repeated 12 times (pre-norm design, `layers/transformers/layers.py`)



Part	Purpose
LayerNorm	Normalises each token for stable training
Self-attention (4 heads)	Every square compares itself with every other square in the patch. A dark debris square can "see" that it connects to a large ice body upstream.
MLP (64 → 256 → 64)	Processes each token after attention has mixed in information. In this code the second Dense layer also uses GELU.
Residual connections (+)	Let gradients pass through 12 blocks without fading
Dropout 0.1	Reduces overfitting

After 12 blocks, the tokens are reshaped to a **24 × 24 × 64** grid.

c) PUP decoder (progressive upsampling) back to full size:

Step	Layer	Output
1	Conv 3×3 + BN	24×24×64
2	Upsample ×2 → Conv 3×3 + BN	48×48×64
3	Upsample ×2 → Conv 3×3 + BN	96×96×64
4	Upsample ×2 → Conv 3×3 + BN	192×192×64
5	Conv 1×1 (linear)	192×192×64
6	Upsample ×2	384×384×64

d) Skip connection: `encoded1 = SETR output + fused`. The SETR output is blurry because it was
built from 160 m squares. Adding the fusion features back restores the 10 m detail.

1.5 Stage 3a — Deep-supervision head

`Dense(2, softmax)` on `encoded1` gives a first, rough glacier map (384×384×2).

- It is used **only in training**. The focal loss is applied with weight 0.5 here and 1.0 on the final output.
- At prediction time it is removed. The code builds a 2-output `train_model` and a 1-output `test_model`.
- **Purpose:** it forces the transformer stage to learn a useful map on its own, and it gives gradients a short path back to early layers.

1.6 Stage 3 — Residual U-Net

Code: `ResUNet(n_steps=3, start_n_filters=64)` in `models/mapping/resunet.py`.

Building block used at every level:

```
x ─┬─► Conv3×3 ─► BN ─► LeakyReLU ─► Conv3×3 ─► BN ─► LeakyReLU ─►
(SpatialDropout) ─► (+) ─► y
   └─► Dense 1×1 (projection, no bias)
──────────────────────────────────────────────────────────────┘
```

Encoder (image gets smaller, channels increase):

Level	Operation	Output	Pixel size
1	Residual block 64 → skip 1	384×384×64	10 m
2	MaxPool, residual block 128 → skip 2	192×192×128	20 m
3	MaxPool, residual block 256 → skip 3	96×96×256	40 m
Bottleneck	MaxPool, residual block 512	48×48×512	80 m

Decoder (image gets larger, channels decrease):

Level	Operation	Output
3	Up-conv to 256, concatenate skip 3, residual block	96×96×256
2	Up-conv to 128, concatenate skip 2, residual block	192×192×128
1	Up-conv to 64, concatenate skip 1, residual block	384×384×64
Out	Dense 64 (linear)	384×384×64

The **skip connections** give the decoder the sharp, high-resolution features of the encoder. This is why U-Nets produce clean boundaries.

A final skip adds the transformer result: `encoded = encoded1 + U–Net output`. The U-Net therefore
learns only a **correction** to the transformer's map.

1.7 Stage 4 — Output layer

`Dense(2, softmax)` applied to every pixel gives **384×384×2**: channel 0 = P(not glacier), channel 1 = P(glacier). The two values add up to 1.

For a full scene, 384 px patches are predicted with a stride of 192 px and blended with Gaussian
weights.

1.8 Layer summary

#	Stage	Main layers	Output	Job
1	Fusion	2 × (Conv + BN + LeakyReLU) and SE gate per group; sum	384×384×64	Turn each data source into features and combine them
2	SETR encoder	Patch embedding, positional embedding, 12	24×24×64	Global context over the whole 3.84 km patch

#	Stage	Main layers	Output	Job
		transformer blocks		
2	SETR decoder	4 × (Conv + BN + upsample)	384×384×64	Back to full size
—	Skip	Add fusion features	384×384×64	Restore fine detail
3a	Deep supervision	Dense(2, softmax)	384×384×2	Training only
3	Residual U-Net	3 levels down, bottleneck, 3 levels up	384×384×64	Sharp 10 m boundaries
—	Skip	Add transformer result	384×384×64	U-Net learns a correction
4	Output	Dense(2, softmax)	384×384×2	P(glacier) per pixel

Size: about 10 million parameters (ViT-Base alone has about 86 million).

1.9 Why this design suits glaciers

Problem	Part of the network that addresses it
Debris-covered ice looks like rock up close	Transformer attention sees the connection to the main ice body
Optical, DEM and SAR are different data types	Separate fusion branches, weighted by SE
Outlines must be accurate to about 10 m	U-Net with skip connections at full resolution
Deep networks are hard to train	Residual connections and deep supervision
Some inputs exist only in some regions	Fusion by addition: branches can be added or removed

1.10 Model variants M0–M3

Model	Learning type	Training labels	Starts from	Inputs	HMA test IoU
M0 global	Supervised (by the authors)	Expert outlines, worldwide	—	6 optical + 2 DEM	0.678
M1 HMA official	Supervised fine-tuning (by the	Expert outlines, HMA	M0	6 optical + 2 DEM	0.748

Model	Learning type	Training labels	Starts from	Inputs	HMA test IoU
	authors)				
M1-local	Supervised fine-tuning (this project)	Expert outlines, 165 HMA tiles	M0	6 optical + 2 DEM	0.758
M2 Chhota Shigri	**Semi-supervised (self-training)**	Pseudo-labels + HMA replay	M1	6 optical + 2 DEM	**0.779**
M3 SAR	Supervised fine-tuning	Expert outlines, 66 HMA tiles with SAR	M0 + new SAR branch	6 optical + 2 DEM + 2 SAR	0.808 †
Optical-only (ablation)	Supervised fine-tuning	Same 66 tiles	M0	6 optical	0.676 †
No-SAR baseline (ablation)	Supervised fine-tuning	Same 66 tiles	M0	6 optical + 2 DEM	0.824 †

† These three are measured on the smaller 22-tile SAR test set and cannot be compared directly with
the 57-tile results. On that set the DEM adds **+0.149 IoU** and SAR gives **−0.016 IoU**.

Input channels

Group	Channels	Why it helps
`optical`	B2 blue, B3 green, B4 red, B8 NIR (10 m); B11 SWIR-1, B12 SWIR-2 (20 m → 10 m)	Ice and snow are bright in visible light and dark in SWIR; clouds are bright in SWIR
`dem`	Elevation (m), slope (°)	Glaciers lie at high altitude on gentler slopes; key for debris-covered ice
`co_pol_sar` (M3)	VV ascending, VV descending	Radar sees through cloud and responds to wet snow and roughness

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2. Error sources and how they are handled

2.1 Principle: the images are never edited

The pipeline **does not remove** snow, rock or shadow from the satellite images. Every Sentinel-2
image goes into GlaViTU unchanged. Obstacles are handled in two ways:

1. **Rejecting** data that cannot be trusted (cloudy images, fresh snowfall, cloudy pixels).
2. **Classifying**: the model, trained on expert labels, decides for each pixel whether it is glacier. Rock, debris, snow and shadow are all things the model learned to separate from ice.

Afterwards, the multi-date median and the post-processing clean up the remaining errors.

> **Key message:** *We do not edit the images. We reject what cannot be trusted, let a model trained*
> *on expert labels classify the rest, and use many images per year so that one-day effects are voted out.*

2.2 The order of the defences

```
 1. Choose the season        avoid seasonal snow, low sun and monsoon cloud
 2. Screen each image        reject cloudy or freshly snowed images
 3. Fix the radiometry       same brightness scale in every year
 4. Align the images         no false change from shifted images
 5. Model prediction         GlaViTU separates glacier from rock, snow,
shadow, debris
 6. Mask cloudy pixels       cloud pixels set to "no data" after prediction
 7. Multi-date median        one-day errors are voted out
 8. Reject outlier images    whole bad images removed
 9. Fill gaps                pixels never seen clearly get a same-year value
10. Post-processing          noise, neighbouring glaciers and small blobs
removed
```

2.3 Atmospheric errors

Cloud — ● addressed

- *Per image* (`sentinel2.py` , `select_ensemble`): a cheap SCL
  pre-screen rejects an image with more than 60 % cloud in the domain. The limit is loose because
  SCL often marks bright ice as cloud. Then s2cloudless (threshold 0.4, 4 px averaging, 2 px = 40 m
  dilation) rejects an image with more than 50 % cloud.
- *Per pixel* (`ensemble.py`): cloudy pixels are set to NaN after
  prediction and take no part in the median.

Cloud shadow — ◐ partly addressed

- The final ensemble has **no per-pixel cloud-shadow mask**. The SCL shadow class is computed but
  used only in an older selection method that is no longer used.
- Partial protection: the 40 m cloud-mask dilation covers shadow next to clouds, and the median and
  outlier rejection reduce the rest.
- Gap: a shadow far from its cloud can still enter the median.

2.4 Surface conditions

Seasonal and fresh snow — ● addressed

- Only images from **15 Aug – 15 Sep** are used, in every year. Most seasonal snow has melted, the
  monsoon is ending, and autumn snowfall has not started.
- An image is rejected if SCL snow covers **more than 45 % of the ground outside the glacier**. Snow
  outside the glacier means recent snowfall.
- This was tested: scenes after 15 September map **1.5–4.6 km² too little glacier**. Extending the
  window to 15 October lowers agreement with the inventory (IoU 0.816 → 0.778) and creates a false
  −2.65 km² step in 2023–2025.

Thin snow on debris margins — ◐ reduced, not solved

- One image shows exposed debris, the next shows a thin snow cover. This alone changes the area by
  about **1 km²**.
- The per-pixel **median over all images of a year** reduces this to 0.3–0.8 km².
- In training, pseudo-labels **ignore** these flickering pixels, so M2 does not learn a wrong answer.
- This is the **largest remaining error** and the main reason area change is not detectable.

Debris-covered ice — ◐ the hardest case

- Debris on ice looks like rock in optical bands.
- Defences: the DEM (shape of the tongue), transformer context, and HMA fine-tuning on debris-rich
  glaciers. The DEM adds +0.149 IoU [0.078, 0.241].

- Remaining: the absolute terminus position differs by up to about 800 m between models, so only
  terminus *change* is reported.

Bare rock and moraine — ● addressed by the model

- No rule removes rock. The model separates rock from ice with the SWIR bands and the DEM.
- Post-processing removes objects smaller than 100 px and fills holes smaller than 50 px.

Surface water (proglacial lakes) — ○ not addressed

- There is no water mask. The model must separate water from ice by itself. This matters if the work
  is framed around GLOF risk.

2.5 Illumination and timing

Terrain shadow — ◐ indirect

- There is no explicit terrain-shadow mask. The hillshade is used only for figures.
- Indirect defences: the window closes before the low-sun period, slope is a model input, and the
  median and outlier rejection reduce shadow effects.

Seasonal timing between years — ◐ partly addressed

- The same calendar window is used every year. However, the actual dates differ: both 2025 scenes
  are from August, while 2023 reaches 11 September. Part of the 2023–2025 change may be seasonal.

2.6 Geometric and radiometric errors

Misregistration — ● addressed (odd years)

- A shift of one pixel moves the glacier edge and looks like retreat or advance.
- Phase correlation on stable terrain corrects shifts larger than 0.25 px (details in 3.8). Two scenes were corrected: 0.67 → 0.11 px and 0.36 → 0.14 px. All residuals are ≤ 1.6 m.
- **The even-year scenes were not co-registered** (see 5. Known issues).

Radiometric baseline change — ● addressed

- From January 2022 (processing baseline 04.00), ESA added a −1000 DN offset to Sentinel-2 data.
- Each product's own `RADIO_ADD_OFFSET` and `QUANTIFICATION_VALUE` are read from its metadata, so
  2017–2021 and 2022–2025 are on the same scale.

2.7 Temporal aggregation

Anomalous whole scenes — ● addressed

- An image is removed if its area differs from the year's median by more than max(3 × 1.4826 × MAD, 0.5 km²). Example: 2022-09-06 was rejected for M1 and M2.

Data gaps — ◐ partly addressed

- A pixel with no clear view in the window is filled from clear images of the **same year** outside the window. Pixels still unseen are reported as unobserved.
- 2017 and 2019 have only 2 scenes each, and about 3 km² of the domain was seen only once.

2.8 Spatial context and noise

Neighbouring glaciers — ● addressed

- The analysis domain is the RGI7 outline + 1 km **minus 33 neighbouring glaciers**.
- Only ice bodies touching the **RGI7 core** (outline shrunk by 300 m) are kept. The core only identifies which ice body is Chhota Shigri; the edges come from the model.

Classification noise — ● addressed

- Threshold 0.5, 3 × 3 median filter, objects < 100 px removed, holes < 50 px filled, slivers < 4 px removed after polygonisation, and validity checks.

2.9 SAR (model M3 only)

Layover and radar shadow — ◐

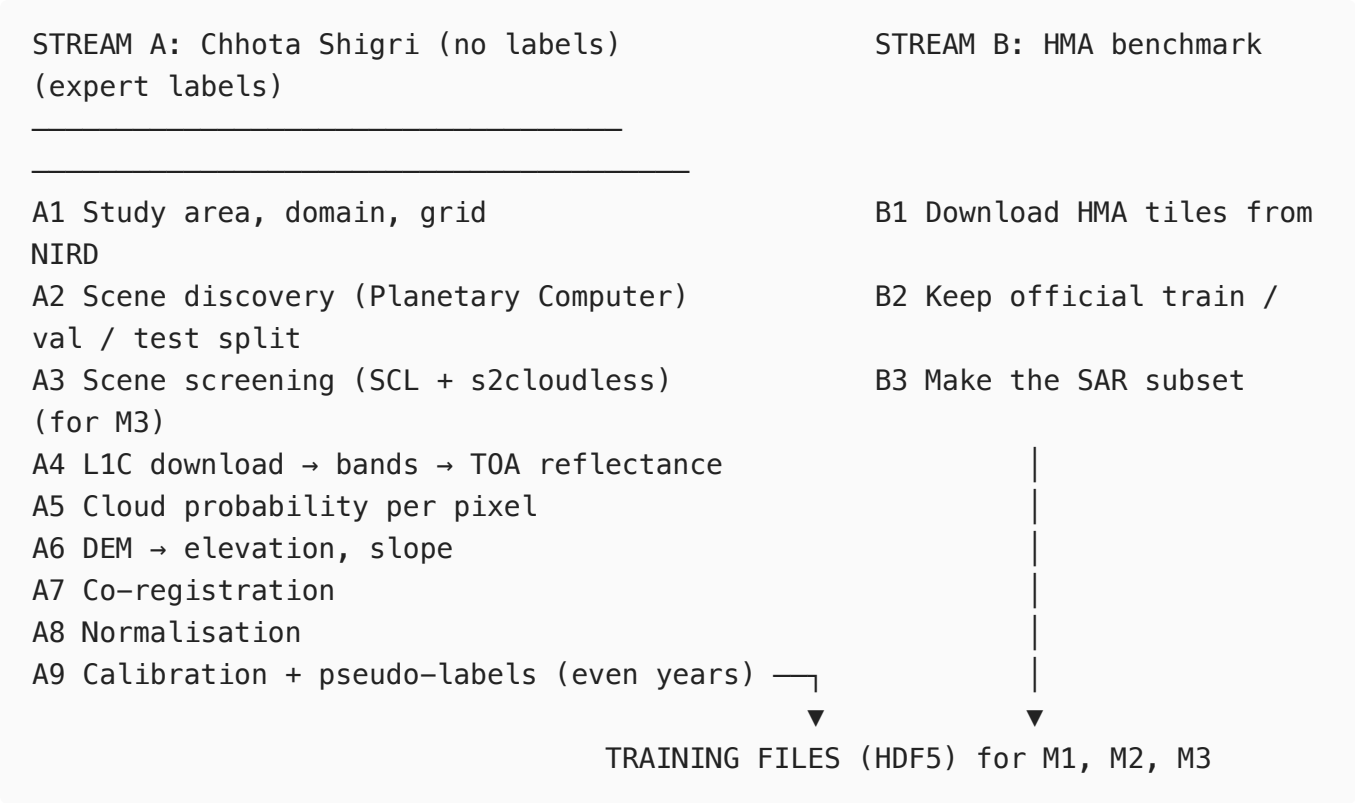
- Missing backscatter on steep slopes is filled with the composite median.
- SAR gave no benefit: −0.016 IoU [−0.029, −0.008] compared with optical + DEM.

2.10 Summary table

Error source	How it is handled	Key parameters	Remaining effect	Status
Cloud	Image rejection + per-pixel NaN	s2cloudless p ≥ 0.4, 40 m dilation, > 50 % → reject	Gaps filled from same year	●
Cloud shadow	Indirect	Dilation, median, outlier rejection	Not quantified	◐
Seasonal / fresh snow	Window + snow screen	15 Aug–15 Sep; > 45 % off-glacier snow → reject	Later scenes: −1.5 to −4.6 km²	●
Thin snow on debris	Median; ignored in pseudo-labels	34.7 % of pixels ignored	~1 km² → 0.3–0.8 km²	◐
Debris-covered ice	DEM + HMA fine-tuning	Elevation, slope	Terminus position spread ~800 m	◐
Bare rock	Model + object filter	< 100 px removed, holes < 50 px	Included in HMA IoU 0.779	●
Proglacial lakes	None	—	Not quantified	○
Terrain shadow	Indirect	Window, slope input, median	Not quantified	◐
Seasonal timing	Fixed window	Same dates every year	2025 scenes August only	◐
Misregistration	Phase correlation	> 0.25 px corrected	Residual ≤ 1.6 m (odd years)	●
Radiometric offset	Harmonisation	RADIO_ADD_OFFSET	None expected	●
Outlier images	MAD rejection	max(3·1.4826·MAD, 0.5 km²)	Logged	●
Data gaps	Same-year gap fill	No mixing of years	~3 km² seen once (2017, 2019)	◐
Neighbouring glaciers	Domain + core connectivity	−33 glaciers; 300 m core	Edges from the model	●
Noise	Morphological filtering	3×3 median, slivers < 4 px	In ½ px × perimeter error	●
SAR layover / shadow	Median fill	—	SAR −0.016 IoU	◐

3. Preprocessing (everything before training)

3.1 Two data streams



3.2 A1 — Study area, domain and grid

Code: `src/data/inventory.py` .
Source: the GLIMS WFS server, inside 77.40–77.75° E, 32.10–32.40° N. It provides RGI 6.0, RGI 7.0
and all GLIMS outlines of Chhota Shigri.

Product	Definition	Purpose
ROI	Bounding box of RGI7 + 2.5 km	Area to download
Grid	UTM 43N, 10 m, **snapped outward to the 60 m lattice** → 1434 × 1254 px	One grid for every product
Domain	RGI7 + 1 km **minus 33 neighbouring RGI7 glaciers** → 47.9 km²	Where glacier pixels are counted
Core	RGI7 shrunk by 300 m	Identifies which ice body is Chhota Shigri

Why it matters:

- **60 m snapping:** Sentinel-2 tile origins are multiples of 60 m, so the 10 m bands align exactly
  with the grid and need no resampling.
- **Fixed domain:** GLIMS outlines range from 11.6 to 16.8 km², mostly because of different ice divides. A fixed domain means area change comes from the margins, not from divide choices.
- **Location from the polygon:** the often-quoted 32.28° N 77.58° E is the Bara Shigri terminus.

3.3 A2 — Scene discovery

Code: `search_l2a` in `src/data/sentinel2.py`.

1. Search Microsoft Planetary Computer (`sentinel-2-l2a`) for all items over the ROI, **15 Aug – 15 Oct**, for each year.
2. Keep only MGRS tile **43SGR** (no duplicates from overlapping tiles).
3. Keep **one item per datatake**, the latest reprocessing.

Why L2A here but L1C later? GlaViTU was trained on L1C (top-of-atmosphere) reflectance, so the
model input must be L1C. Planetary Computer serves only L2A, but L2A has the SCL layer, which is
useful for screening. L2A is therefore used **only to find and screen** scenes.

3.4 A3 — Scene screening

Gate	Test	Reject if	Error source
1	SCL cloud (classes 8, 9, 10) over the domain	> 60 %	Cloud (cheap first pass)
2	SCL snow (class 11) on **domain minus glacier**	> 45 %	Fresh snowfall
3	s2cloudless cloud over the domain (20 m)	> 50 %	Cloud (accurate test)

Each decision and reason is saved to `data/sentinel2/ensemble_<year>.csv`. The **core window**
15 Aug – 15 Sep is applied when scenes are used; later scenes serve only for gap filling and the
sensitivity test.

3.5 A4 — L1C download, band selection, radiometry

Code: `find_l1c` and `fetch_l1c_bands` in `src/data/sentinel2.py`.
Source: Google Cloud public Sentinel-2 bucket (anonymous HTTPS).

1. Find the L1C product with the same datatake; prefer the highest processing baseline.
2. **Select 6 bands**, exactly those GlaViTU was trained with:

Band	Native resolution	Resampling to 10 m
B2, B3, B4, B8	10 m	None (grid aligned)
B11, B12	20 m	Bilinear

3. Read **only the grid window** over HTTP (GDAL `/vsicurl`). Data are already in UTM 43N, so no
reprojection is needed.
4. **Convert to TOA reflectance:**
`reflectance = (DN + RADIO_ADD_OFFSET) / QUANTIFICATION_VALUE`, with both values read from each
product's `MTD_MSIL1C.xml`. This handles the 2022 offset change.
5. DN = 0 → NaN (no data).
6. Save `data/sentinel2/scenes/<date>/optical_toa.tif` and `scene.json`.

3.6 A5 — Cloud probability

1. Read 10 L1C bands at 20 m (B1, B2, B4, B5, B8, B8A, B9, B10, B11, B12).
2. Run s2cloudless: threshold 0.4, 4 px averaging, 2 px (40 m) dilation.
3. Upsample to 10 m and save `cloud_prob.tif`.

Used for the image-level test above, and later to give cloudy pixels **no label** in training.

3.7 A6 — DEM: elevation and slope

Code: `src/data/dem.py`. Source: Copernicus DEM **GLO-30** (Planetary
Computer); heights relative to EGM2008.

1. Mosaic all DEM tiles over the ROI.
2. **Elevation:** reproject to the 10 m grid (bilinear).
3. **Slope:** reproject to 30 m (cubic), calculate with **Horn's (1981) method**, upsample to 10 m (bilinear).
4. Aspect and hillshade are also made, for analysis and figures only.

Why slope at 30 m? The DEM's real resolution is 30 m. Slope from a DEM interpolated to 10 m has
artificial steps.

3.8 A7 — Co-registration

Code: `src/preprocessing/coregister.py`.

1. Reference image: 2023-09-11.
2. **Stable terrain** only: outside the domain + 300 m, outside all glaciers + 150 m, slope < 35°, cloud-free in both images, snow-free in both images (NDSI < 0.2).
3. Convert band B8 to Sobel gradient magnitude (edges match better than raw brightness).
4. Phase cross-correlation (×20 upsampling, sub-pixel) in 256 px windows with ≥ 50 % stable pixels.
5. If the median shift is > 0.25 px, apply a Fourier shift to all bands and the cloud probability.

Year	Shift before (m)	Residual after (m)	Scenes corrected
2017	6.67	1.41	1 (0.67 → 0.11 px)
2019	3.61	1.41	1 (0.36 → 0.14 px)
2021	0.71	0.71	0
2023	1.41	1.41	0
2025	1.58	1.58	0

Even-year scenes were not processed by this step.

3.9 A8 — Normalisation

Code: `normalise` in `src/inference/glavitu.py`.

1. Optical NaN → 0.
2. **Min-max scaling with the official global statistics** (`dataset_stats.pickle`), separately for
   each optical band, elevation and slope: `x_norm = (x − min) / (max − min)`.
3. Group into `optical` (6 channels) and `dem` (2 channels).

Why the authors' statistics? The pretrained weights expect exactly the scale of the global benchmark. Statistics from Chhota Shigri would shift every input and damage the pretrained model.

3.10 A9 — Calibration and pseudo-labels

This is the last step before M2 training. Code:
`src/validation/calibration.py` and
`src/training/pseudo_labels.py`.

What is a pseudo-label? A label made by a model instead of a person. The **teacher** (M1) makes
labels; the **student** (M2) trains on them. There is no hand-drawn outline of Chhota Shigri, so this
is the only way to train on the site without breaking the rule "no manual digitising".

Step 1 — Teacher predictions. M1 runs on the even-year scenes of the core window. Cloudy pixels → NaN.

Year	Scenes	Dates
2018	3	18 Aug, 28 Aug, 12 Sep
2020	4	17 Aug, 22 Aug, 1 Sep, 11 Sep
2022	5	22 Aug, 27 Aug, 1 Sep, 6 Sep, 11 Sep
2024	3	21 Aug, 31 Aug (empty, see 5), 5 Sep

Step 2 — Calibration. The raw GlaViTU output never leaves about **0.24–0.89**, because of label smoothing (0.1) and focal loss in training. The rule "P ≥ 0.95" therefore selected **0 pixels** at the first attempt.

Calibration corrects a model's numbers so that "P = 0.95" really means "correct 95 % of the time".
Isotonic regression learns this correction as a step curve that only goes up, so it keeps the model's ranking of pixels. It was fitted on 400,000 pixels from 20 HMA **validation** tiles (true labels; never site data) and tested on a separate half of the tiles.

Raw P	**0.25**	**0.30**	**0.40**	**0.50**	**0.60**	**0.70**	**0.80**	**0.85**
Calibrated P	0.00	0.06	0.33	0.63	0.86	**0.97**	0.999	1.00

The calibration error (ECE) dropped from **0.239 to 0.011**. Raw 0.70 already means 97 % correct.

Step 3 — Consensus rule. A pixel is labelled **glacier** only if all of these are true:

1. Calibrated P ≥ 0.95 in at least 60 % of its clear scenes.
2. P is never ≤ 0.05 in any scene.
3. The yearly median is ≥ 0.95 in at least 3 of the 4 years.

Background uses the opposite rule. **All other pixels are ignored.**

Class	Result
Glacier	292,012 px
Background	881,732 px
Ignored	34.7 % (margins, terminus, flickering snow, real change; 3,999 px flip between scenes)

Step 4 — Training file `data/labels/cs_pseudolabels.hdf5` (same layout as the benchmark):

Per scene, group `CS-<date>`	Shape	Content
`optical`	1254 × 1434 × 6	Normalised reflectance of that scene
`dem`	1254 × 1434 × 2	Normalised elevation and slope
`outlines`	1254 × 1434 × 2	[1,0] background, [0,1] glacier, **[0,0] ignored** (zero loss)

- The **same consensus label** is attached to every scene, so the student learns not to be fooled by
  one-day snow or shadow.
- A pixel that is cloudy in a scene gets [0,0] in that scene.
- Each scene appears **twice** (`CS-<date>` and `CS-<date>-r1`) to raise its sampling weight.
- **40 HMA training tiles** are linked in for replay, so the model does not forget expert labels.
- Total: 70 entries (30 site + 40 HMA).
- No inventory outline is used as a label, and the M2 checkpoint is selected on **HMA validation**
  tiles, never site data.

Why does a student learn from its teacher's own answers?

1. The consensus over many scenes and years is better than any single prediction.
2. The student learns local conditions (snow, debris, shadow) of Chhota Shigri.
3. The difficult margins get no site label, so the student cannot copy the teacher's errors there.

Result: HMA test IoU 0.748 → 0.779; agreement with RGI7 0.816 → 0.838; boundary distance 61 → 48 m.

3.11 B — HMA benchmark tiles

Code: `src/data/benchmark.py` . Source: the GlaViTU benchmark on the NIRD archive (doi:10.11582/2024.00168), 411 GB in total.

1. Read **only the HMA tiles** with HTTP byte-range requests (no full download).
2. Four parallel shards; each tile is written as `.tmp` and renamed when complete.
3. Keep `optical` , `dem` , `outlines` , `co_pol_sar` , `cross_pol_sar` as provided by the authors.
4. Keep the **official split: 165 train / 55 val / 57 test.** No tile lies within 0.05° of the site.
5. SAR subset (tiles with `co_pol_sar`): 66 train / 22 val / 22 test.

3.12 Sentinel-1 (special case)

Code: `src/data/sentinel1.py` . Site SAR is **not used in any training**; it is used only when M3 is applied to the site.

- Source: Planetary Computer `sentinel-1-rtc` (γ^0).
- Approximate $\sigma^0 = \gamma^0 \times \cos(\text{incidence angle})$, incidence about 37–39°.
- Per-pixel median per orbit direction in the core window; gaps filled with the median.
- Normalised as log10(x + 1e-6), then official min-max.
- Limitation: terrain flattening is not undone, so this is an approximate input.

3.13 Training files for each model

Model	Training file	Validation file	Inputs
M1-local	`hma_train.hdf5` (165 tiles)	`hma_val.hdf5` (55)	optical + dem
M2	`cs_pseudolabels.hdf5` (15 scenes × 2 + 40 HMA)	`hma_val.hdf5`	optical + dem
M3	`hma_train_sar.hdf5` (66)	`hma_val_sar.hdf5` (22)	optical + dem + SAR
No-SAR / optical-only	`hma_train_sar.hdf5` (66)	`hma_val_sar.hdf5` (22)	optical + dem / optical

Preprocessing ends here. During training, the official loader cuts random 384 px patches and applies
augmentations on the fly (flips, rotation, crop-and-scale, gamma 0.8–1.2, feature occlusion, Gaussian
shift and noise, label smoothing 0.1).

3.14 Error sources handled before training

Step	Error source handled
A1 Domain and grid	Neighbouring glaciers; differing ice divides; band alignment
A2 Discovery	Duplicate scenes
A3 Screening	Clouds; fresh snowfall
Core window	Seasonal snow; autumn snowfall; low sun
A4 Radiometry	ESA 2022 offset
A5 Cloud probability	Per-pixel clouds (and nearby shadow)
A6 DEM	Debris-covered ice vs rock; slope artefacts
A7 Co-registration	Image shifts (odd years)
A8 Normalisation	Scale mismatch with the pretrained model
A9 Pseudo-labels	Cloudy pixels unlabelled; flickering margins ignored; temporal leakage
B Benchmark	Spatial leakage (official split, no tile near the site)

Not handled before training: per-pixel cloud-shadow mask, terrain-shadow mask, water mask.

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4. Analysis of all years (baseline 2017)

4.1 How to read the tables

- **Baseline:** every change is relative to **2017**.
- **Sign convention:** ΔA negative = glacier area lost. Retreat positive = terminus moved upstream.
- **Odd years** are the analysis epochs. **Even years** were also mapped by every model (from 11 October 2026). They add four extra points to the time series.
- **†** marks M2 in an even year. M2 was trained on exactly those scenes, so its values there are
  not independent. The proof: M2's between-scene spread in 2018 is 0.04 km², against 0.55 km²
  for M1 on the same scenes. For independent even-year values, use **M0 or M1-official**.
- **‡** marks even years, which were not co-registered.
- Each yearly product is a multi-date ensemble (per-pixel median) of the core-window scenes.

- Full LaTeX versions: `docs/paper/table_allyears_baseline2017.tex` .
  Data: `docs/paper/data/` .

4.2 Data and quality of each year

	2017	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Role	baseline	training	analysis	training	analysis	training	analysis	training
Candidate scenes	3	6	6	6	6	6	6	6
Rejected: cloud / snow	1 / 0	3 / 0	4 / 0	2 / 0	4 / 0	1 / 0	0 / 0	3 / 0
Rejected: empty / outlier	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 1	0 / 0	1 / 0
Scenes used	**2**	**3**	**2**	**4**	**2**	**4**	**6**	**2**
Platforms	2 S2B	2 S2B, S2A	2 S2A	S2B, 3 S2A	2 S2A	3 S2A, S2B	3 S2A, 3 S2B	S2A, S2B
Sun elevation (°)	58–64	58–65	60–62	58–65	60–65	59–64	59–65	60–64
Co-registration residual (m)	1.41	‡	1.41	‡	0.71	‡	1.41	‡
Mean clear looks / pixel	1.90	2.79	1.91	3.91	1.96	3.82	5.27	1.88
Area seen once (km²)	3.41	0.57	3.14	0.00	1.68	0.03	0.00	5.64
Gap-filled (km²)	0.10	0.00	0.52	0.00	0.00	0.00	0.00	0.14
Unobserved (km²)	0.52	0.15	0.01	0.00	0.00	0.00	0.00	0.00

Reading: 2023 is the best-observed year (6 scenes, every pixel seen about 5 times). 2017, 2019
and 2024 are the weakest (2 scenes, 3–6 km² seen only once).

4.3 M2 area and uncertainty

	2017	2018†	2019	2020†	2021	2022†	2023	2024†	2025
Area (km²)	**15.00**	14.76	**15.89**	15.11	**15.38**	14.96	**15.03**	14.69	**14.30**
Perimeter (km)	80.9	79.3	85.6	81.1	82.1	77.9	82.8	82.2	79.6
σ buffer	0.40	0.40	0.43	0.41	0.41	0.39	0.41	0.41	0.40
σ threshold	1.12	0.79	1.12	0.87	0.91	0.74	0.88	0.81	0.75
σ surface	0.81	0.04	0.75	0.07	0.32	0.13	0.37	0.37	0.27
σ total	**1.44**	0.88	**1.41**	0.96	**1.05**	0.85	**1.04**	0.98	**0.89**
Single-scene areas	14.55–15.70	14.53–14.86	15.27–16.32	15.11–15.47	15.07–15.53	14.84–15.05	14.50–15.67	14.38–14.90	14.12–14.50

The three uncertainty terms:

- **σ buffer** = ½ pixel × perimeter (edge position).
- **σ threshold** = half the area difference between thresholds 0.4 and 0.6 (calibrated probability).
- **σ surface** = spread between the scenes of a year (snow and debris conditions).

σ total = √(σ buffer² + σ threshold² + σ surface²). Published area: 15.5–15.7 km².

4.4 M2 change relative to 2017

	2018†	2019	2020†	2021	2022†	2023	2024†	2025
ΔA (km²)	−0.24	+0.89	+0.12	+0.38	−0.04	+0.03	−0.30	**−0.70**
σ ΔA (km²)	1.05	1.25	1.03	1.07	1.07	1.09	1.11	1.09
ΔA (%)	−1.6	+6.0	+0.8	+2.5	−0.2	+0.2	−2.0	−4.7
Rate since 2017 (km² a^{-1})	−0.25	+0.45	+0.04	+0.10	−0.01	+0.01	−0.04	−0.09
Significant at 2σ	no	no	no	no	no	no	no	no
Pixels lost since 2017 (km²)	0.61	0.50	0.49	0.51	0.54	0.66	0.74	**0.94**
Pixels gained since 2017 (km²)	0.37	1.40	0.60	0.89	0.51	0.70	0.44	0.24
Terminus position (m along centerline)	9770	10090	9850	9850	9740	9850	9830	9740

	2018†	2019	2020†	2021	2022†	2023	2024†	2025
Retreat, centerline (m)	+110	−210	+30	+30	+140	+30	+50	**+140**
Retreat, box method (m)	+83	−218	+47	+35	+111	+41	+49	**+108**
σ retreat (m)	14.2‡	14.3	14.2‡	14.2	14.2‡	14.3	14.2‡	14.3

How change uncertainty is calculated: the buffer and surface terms of both years are combined in
quadrature; the threshold term is treated as systematic (the same model shifts every year the same
way).

Reading:

- No area change relative to 2017 is statistically significant.
- Lost area grows steadily from 0.5 to 0.94 km², while gained area falls from 1.4 to 0.24 km². The
  outline is slowly shrinking even though the net change is within the noise.
- The 2019 "advance" (−210 m) and "gain" (+1.40 km²) come from a weak epoch with 2 scenes and a large
  co-registration correction. They are not physical.

4.5 All models: area and change

Area (km²)

Model	2017	2018	2019	2020	2021	2022	2023	2024	2025
M0 global	13.19	13.42	14.47	15.15	14.70	13.96	14.44	13.32	13.42
M1 HMA official	14.16	13.96	14.81	14.54	14.78	14.41	14.46	13.95	13.79
M1 HMA local	14.42	—	15.43	—	15.24	—	15.26	—	14.10
M2 Chhota Shigri	15.00	14.76†	15.89	15.11†	15.38	14.96†	15.03	14.69†	14.30
M3 HMA + SAR	14.38	14.83	16.05	15.62	15.72	—	15.53	14.29	13.96

Model	2017	2018	2019	2020	2021	2022	2023	2024	2025
Median (range)	14.38 (1.80)	14.36 (1.40)	15.43 (1.59)	15.13 (1.09)	15.24 (1.03)	14.41 (1.00)	15.03 (1.09)	14.12 (1.38)	13.96 (0.88)

ΔA relative to 2017 (km²)

Model	2018	2019	2020	2021	2022	2023	2024	2025
M0 global	+0.23	+1.28	+1.96	+1.51	+0.76	+1.24	+0.13	+0.22
M1 HMA official	−0.20	+0.65	+0.38	+0.62	+0.25	+0.30	−0.21	−0.37
M1 HMA local	—	+1.01	—	+0.82	—	+0.84	—	−0.32
M2 Chhota Shigri	−0.24†	+0.89	+0.12†	+0.38	−0.04†	+0.03	−0.30†	−0.70
M3 HMA + SAR	+0.44	+1.67	+1.24	+1.34	—	+1.14	−0.10	−0.43
Median	+0.01	+1.01	+0.81	+0.82	+0.25	+0.84	−0.15	−0.37

M1-local was run on odd years only. M3 has no 2022 result (no Sentinel-1 composite).

Reading: the 2017 baseline is low for most models (M0 especially), so most years show a positive
ΔA. This is a weakness of the 2017 epoch, not glacier growth. From 2023 to 2025 every model shows a
decrease.

4.6 Terminus retreat relative to 2017

Two methods are used:

- **Centerline:** the furthest point downstream reached continuously from the glacier head along an
  automatic least-cost centerline (after Kienholz et al. 2014).
- **Box method:** the glacier area inside a 2 km × 159 m box over the tongue, divided by the box width.

Centerline (m)

Model	2018	2019	2020	2021	2022	2023	2024	2025
M0 global	370	440	160	330	470	370	90	460
M1 HMA official	240	0	150	160	270	160	100	230

Model	2018	2019	2020	2021	2022	2023	2024	2025
M1 HMA local	—	−110	—	−90	—	−90	—	10
M2 Chhota Shigri	110†	−210	30†	30	140†	30	50†	140
M3 HMA + SAR	50	−120	30	40	—	50	100	150
Median [min, max]	175 [50, 370]	−110 [−210, 440]	90 [30, 160]	40 [−90, 330]	270 [140, 470]	50 [−90, 370]	95 [50, 100]	**150 [10, 460]**

Box method (m)

Model	2018	2019	2020	2021	2022	2023	2024	2025
M0 global	386	465	203	338	501	350	144	498
M1 HMA official	250	71	147	176	290	161	110	260
M1 HMA local	—	−100	—	−56	—	−51	—	17
M2 Chhota Shigri	83†	−218	47†	35	111†	41	49†	108
M3 HMA + SAR	65	−126	31	47	—	47	97	127
Median [min, max]	167 [65, 386]	−100 [−218, 465]	97 [31, 203]	47 [−56, 338]	290 [111, 501]	47 [−51, 350]	104 [49, 144]	**127 [17, 498]**
All available models retreat	yes	no	yes	no	yes	no	yes	yes
Models available	4	5	4	5	3	5	4	5

Measurement σ ≈ 14.3 m (odd years) and ≥ 14.2 m (even years, underestimated).

Reading: by 2025 every model places the terminus upstream of its 2017 position (box method: 17–498 m). The "all models retreat" row cannot be compared between odd and even years, because
M1-local, the only model showing an advance, exists only in odd years.

Period results (odd years, 5 models) — the most robust terminus result of the project:

Period	Centerline: median [min–max]	Box: median [min–max]	All 5 models retreat
2017–2019	−110 [−210, 440]	−100 [−218, 465]	no
2019–2021	160 [−110, 240]	105 [−127, 252]	no
2021–2023	0 [0, 40]	5 [−14, 12]	no
2023–2025	100 [70, 110]	80 [67, 148]	yes
2021–2025	**110 [70, 130] (≈ 28 m a^{-1})**	**80 [73, 160] (≈ 20 m a^{-1})**	**yes**
2017–2025	150 [10, 460]	127 [17, 498]	yes

4.7 Agreement with the RGI7 outline

RGI7 is a 2002 outline, so these numbers measure **agreement, not accuracy**. A slow decline over time
is expected if the glacier shrinks.

	2017	2018	2019	2020	2021	2022	2023	2024	2025
M2 IoU	0.836	0.845†	0.833	0.851†	0.854	0.848†	0.843	0.839†	0.823
M2 precision	0.965	0.978	0.934	0.970	0.963	0.974	0.968	0.977	0.981
M2 recall	0.862	0.861	0.885	0.874	0.883	0.868	0.867	0.856	0.836
M2 mean boundary distance (m)	48.9	44.2	52.2	41.8	42.6	43.7	45.0	45.9	51.0
M2 P95 boundary distance (m)	165.5	160.0	200.0	152.3	160.0	153.0	156.2	155.2	170.9
M1 official IoU	0.800	0.808	0.827	0.830	0.833	0.820	0.822	0.804	0.796
M1 mean boundary distance (m)	60.3	61.3	58.8	52.6	58.0	60.0	57.7	62.1	67.9

Reading: precision is high (≥ 0.93) in every year, so the model rarely marks non-glacier as glacier. Recall falls after 2021 (0.883 → 0.836), which fits a glacier that is smaller than its 2002 outline.

4.8 Area trends

Weighted least-squares trend of area against time, with 95 % confidence interval (km² a^{-1}):

Model	Odd years only	All available years
M0 global	−0.089 [−0.456, 0.277]	−0.145 [−0.384, 0.094]
M1 HMA official	−0.180 [−0.384, 0.024]	−0.083 [−0.209, 0.043]
M1 HMA local	−0.024 [−0.317, 0.269]	(odd years only)
M2 Chhota Shigri	−0.159 [−0.402, 0.084]	−0.064 [−0.181, 0.052]
M3 HMA + SAR	−0.114 [−0.611, 0.382]	−0.098 [−0.342, 0.145]

Every confidence interval includes zero. The long-term published loss (≈ 0.03 km² a^{-1}) implies
only about 0.25 km² over 2017–2025, which is below the detection limit of this data.

4.9 Key findings

1. **Terminus retreat is detected.** Over 2021–2025, all five models agree: 70–160 m, about 20–28 m a^{-1}. This is far larger than the ≈ 14 m measurement error.
2. **Relative to 2017, the terminus is upstream in 2025 for every model** (box method 17–498 m).
3. **Area change is not detectable.** No change relative to 2017 and no trend is significant at 95 %.
   The uncertainty (≈ 1 km²) is larger than the expected change (≈ 0.25 km²).
4. **The glacier outline is slowly shrinking.** M2 pixels lost since 2017 grow from 0.5 to 0.94 km², while pixels gained fall from 1.4 to 0.24 km².
5. **Surface conditions are the main noise source.** Thin snow on debris margins changes single-image
   area by about 1 km²; the multi-date median reduces this to 0.3–0.8 km².
6. **The best-observed period is 2021–2025.** The early years (2017, 2019) have only 2 images each and
   needed the largest co-registration corrections.

4.10 Limitations of the year-by-year analysis

- **2017 is a weak baseline**: 2 scenes, 3.4 km² seen once, the largest co-registration correction.
  Every "relative to 2017" number carries this weakness.
- **M2 even-year values are not independent** (†). Use M0 or M1-official for even years.
- **Even years were not co-registered** (‡); their terminus σ is underestimated.
- **The model set differs by year** (M1-local odd years only; M3 has no 2022).
- **Seasonal timing differs**: 2025 has only August scenes; 2023 reaches 11 September.
- **Absolute terminus position** differs by up to ~800 m between models (debris-covered snout); only
  changes are meaningful.

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5. Known issues and corrections

These were found while checking the data for this report. Correct them before publishing.

#	Issue	Evidence	Action
1	`docs/methodology.md` says site scenes were "hard-linked 6 times" in the M2 training file.	The file holds each scene **twice** (`CS-<date>` , `CS-<date>-r1`); code default `--cs_repeats 2` .	Write "2×" ir paper; corre `methodolo`
2	Scene 2024-08-31 is an **empty L1C download** (56 KB instead of ~24 MB).	Dropped by the ensemble; 0 labelled pixels in the pseudo-label file.	M2 trained even-year s with data, n Correct the download th
3	`README.md` says 2017 "was mostly Sentinel-2A only".	Both 2017 scenes used are **Sentinel-2B**.	Correct the README.
4	Even-year scenes were **not co-registered**.	`outputs/statistics/coregistration.csv` lists odd-year scenes only.	Run `coregister` even-year s before using year terminu results.
5	`outputs/statistics/` contains odd years only.	Even-year ensembles were produced on 11 October, after the statistics run of 3 October.	All-year stat in `docs/paper` re-run `chan` needed.

#	Issue	Evidence	Action
6	No per-pixel cloud-shadow, terrain-shadow or water mask.	`src/inference/ensemble.py` masks only s2cloudless cloud and missing data.	State as a li (rows ◐ / ○

Appendix

A.1 Glossary

Term	Meaning
Ablation season	The part of the year when the glacier loses mass by melting (summer)
Box method	Terminus measure: glacier area inside a fixed box over the tongue, divided by the box width
Calibration	Correcting model probabilities so that "P = 0.95" means "correct 95 % of the time"
Centerline	A line along the glacier flow, here computed automatically on the DEM
Co-registration	Aligning images so that the same ground point has the same pixel in every image
Deep supervision	An extra output inside the network that receives its own loss during training
DEM	Digital elevation model (terrain heights)
ECE	Expected calibration error: average gap between predicted probability and real frequency
Ensemble (multi-date)	Combining many images of a year by taking the per-pixel median
Focal loss	A loss that focuses training on difficult pixels
HMA	High Mountain Asia, the region of the benchmark tiles
IoU	Intersection over union: overlap between prediction and reference, divided by their union
Isotonic regression	A calibration method that learns a correction curve that only goes up
L1C / L2A	Sentinel-2 products: top-of-atmosphere (L1C) and surface reflectance with scene classification (L2A)
MAD	Median absolute deviation, a robust measure of spread

Term	Meaning
NDSI	Normalised difference snow index, (green − SWIR) / (green + SWIR)
Pseudo-label	A training label made by a model instead of a person
Replay	Mixing old training data into new training so the model does not forget
RGI	Randolph Glacier Inventory (global glacier outlines)
s2cloudless	A pretrained cloud detector for Sentinel-2
SCL	Scene classification layer of Sentinel-2 L2A (cloud, shadow, snow, …)
Self-attention	The transformer operation in which every token compares itself with every other token
Semantic segmentation	Giving a class to every pixel of an image
Squeeze-and-excitation	A block that learns a weight for each feature channel
TOA reflectance	Top-of-atmosphere reflectance (no atmospheric correction)
WLS	Weighted least squares, a trend fit that gives precise points more weight

A.2 File reference

Topic	Files
Pipeline	`run_pipeline.sh`, `configs/chhota_shigri.yaml`, `configs/hma_finetune.yaml`
Model code	`third_party/glavitu/models/mapping/glavitu.py`, `setr.py`, `resunet.py`; `src/inference/glavitu.py`
Data	`src/data/inventory.py`, `sentinel2.py`, `dem.py`, `sentinel1.py`, `benchmark.py`
Preprocessing	`src/preprocessing/coregister.py`
Training	`src/training/finetune.py`, `pseudo_labels.py`, `src/validation/calibration.py`
Inference	`src/inference/ensemble.py`, `src/postprocessing/mask.py`
Change	`src/change_detection/change.py`, `centerline.py`, `patterns.py`
Results	`outputs/statistics/`, `experiments/<model>/<year>/summary.json`, `docs/paper/data/`
Paper tables	`table_error_sources.tex`, `table_epochs.tex`, `table_allyears_baseline2017.tex`

Topic	Files
Other docs	`README.md`, `methodology.md`, `validation.md`, `results.md`, `experiments.md`